

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B. Tech. Branch : Mechanical & Allied Semester : VII Subject Code & Name: BTMEC701 Mechatronics Max Marks: 60 Date: 1/7/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define Mechatronics and explain the elements of mechatronics system in detail	CO 1	6
B)	Define sensors and transducers and explain working principal of Optical encoder	CO 1	6
C)	Explain Proportional Control, Integral control, Derivative Control with suitable example	CO 5	6
Q.2	Solve Any Two of the following.		12
A)	What is an operational amplifier? Describe summing operational amplifier using block diagram and derive equation for output	CO 2	6
B)	Describe closed loop control system with suitable example	CO 5	6
C)	Explain A - D conversion process using any one type of ADC device	CO 2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain working principal of stepper motors with neat figure and applications	CO 3	6
B)	How cylinder sequencing of pneumatic system is achieved using cascading method? Explain with suitable example	CO 3	6
C)	Describe architecture of PLC using block diagram	CO 4	6
Q.4	Solve Any Two of the following.		12
A)	Explain various pins and pin function in 8081 microcontroller using pin diagram	CO 4	6
B)	Explain function and importance of FRL unit in pneumatic systems using appropriate figure	CO 3	6
C)	Write and explain PLC ladder program for sequential control of three motors M1, M2, M3 in an industrial application where each motor is to be operated with a time delay of 5s after motor M1 starts.	CO 4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain working principal of Thermocouple with neat sketch and applications	CO 1	6

B)	Explain need for signal conditioning with steps involved in signal conditioning	CO 2	6
C)	Determine transfer function $C(s)/R(s)$ for the system shown in below diagram using block diagram reduction method	CO 5	6

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